

MQTT and Node-Red Setups Document

Project: Connected Solar Tracker using IoT (MQTT + Node-RED)

Team: SolarLink7 (THM / ARRAS)

1. Software Used

Software	Purpose	Link
MQTT Explorer	Monitoring MQTT messages	https://mqtt-explorer.com/
Node js	To install Node-Red to work	https://nodejs.org/en/download
Node - Red	Dashboard and Visualization	Node-RED

2. MQTT Configuration

Click on (+) and add new connections with following parameters.

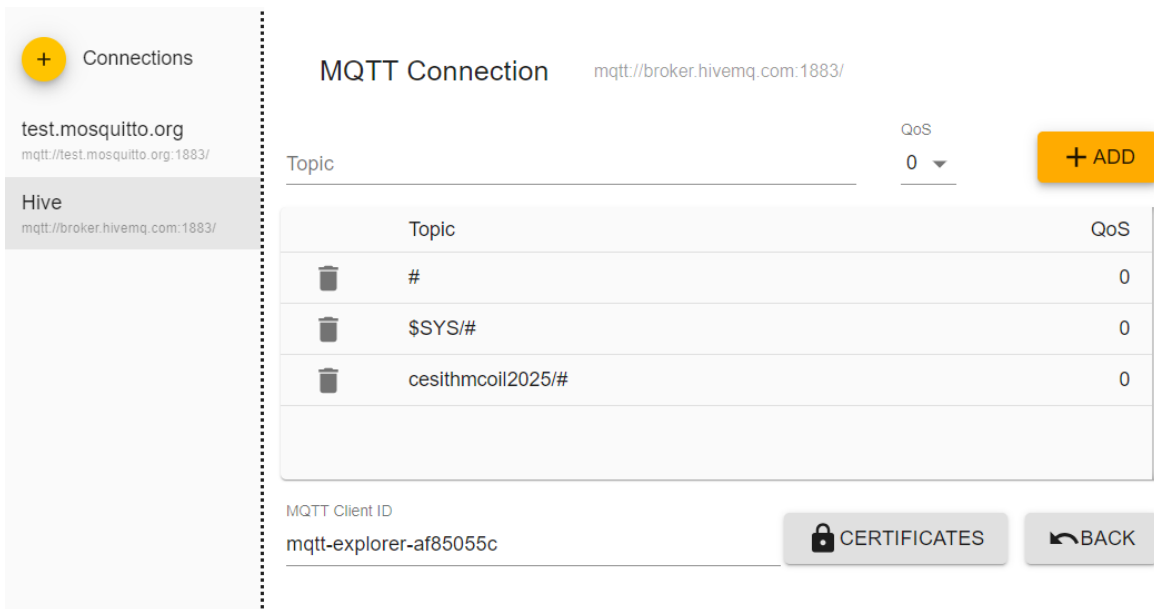
Name: Hive

Host: Broker: broker.hivemq.com

Port: 1883

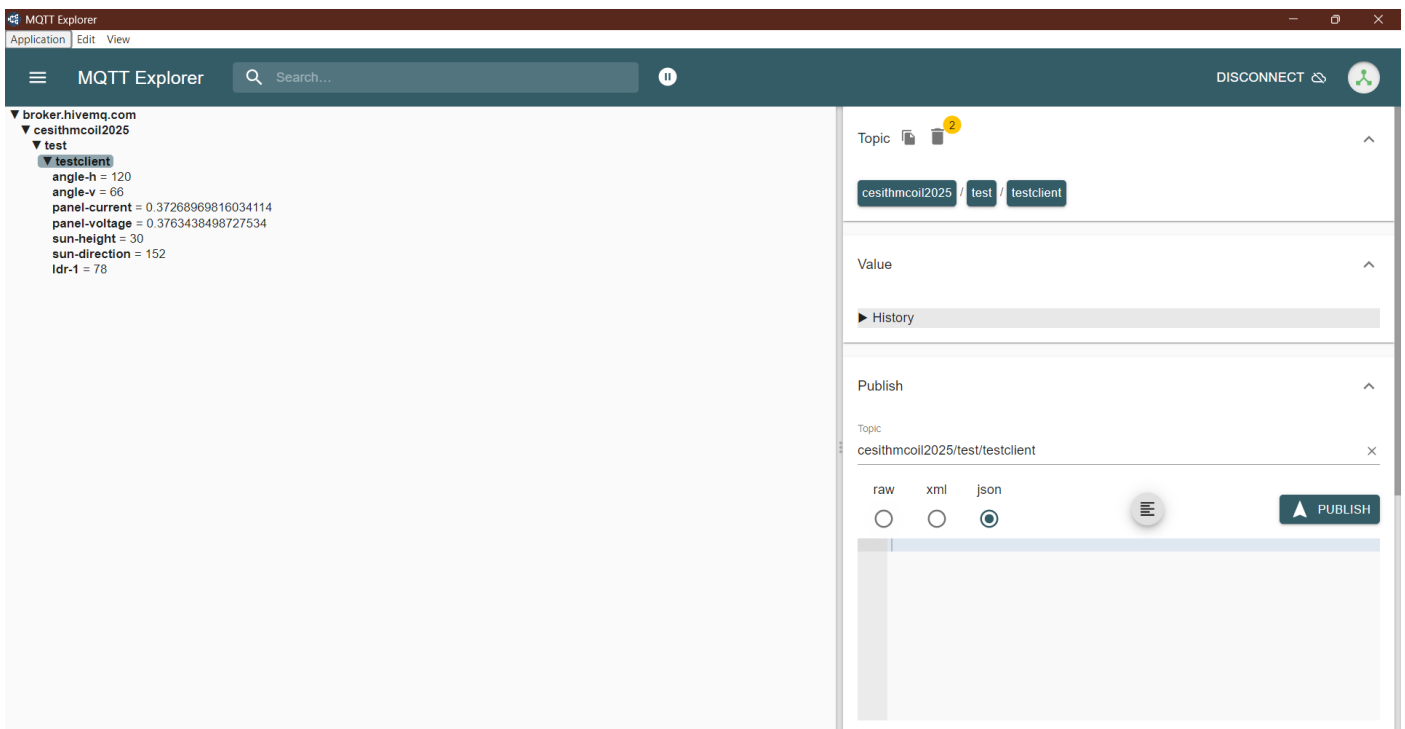
The screenshot shows the MQTT Explorer application interface. On the left, there is a sidebar with a 'Connections' section containing a list of connections: 'test.mosquitto.org' and 'Hive'. The 'Hive' connection is selected. The main area displays the configuration for the 'Hive' connection. At the top, it says 'MQTT Connection' followed by the URL 'mqtt://broker.hivemq.com:1883/'. Below this, there are several fields and controls: 'Name' is set to 'Hive'; 'Validate certificate' is a toggle switch that is turned on; 'Encryption (tls)' is a toggle switch that is turned off. Below these, there are three fields: 'Protocol' is set to 'mqtt://', 'Host' is set to 'broker.hivemq.com', and 'Port' is set to '1883'. At the bottom, there are fields for 'Username' and 'Password'. At the very bottom, there are four buttons: 'DELETE' (with a trash icon), 'ADVANCED' (with a gear icon), 'SAVE' (in a yellow box with a floppy disk icon), and 'CONNECT' (in a dark blue box with a power icon).

Then click on **Advanced**



and in **Topic** add cesithmcoil2025/# and click **+ADD**, then **BACK** and press **Connect**

After 5 sec you will see following if you go on clicking the subtopics (this is just the test Data)

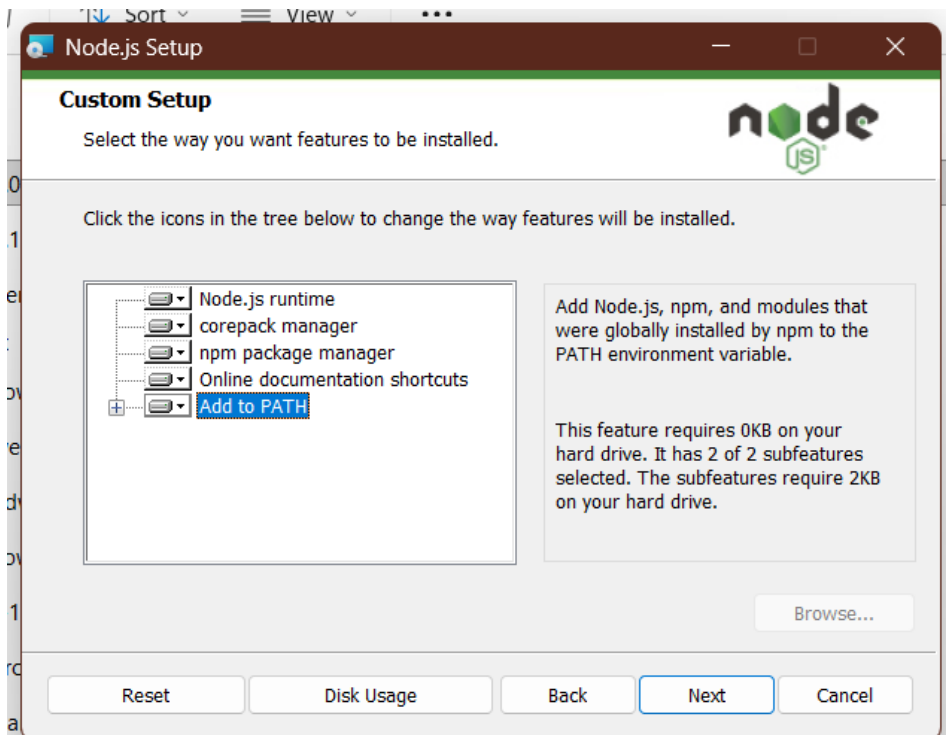


Successful connection to **MQTT broker**.

3. Node- Red Configuration

Install first **node js** from the link.

Make sure to click the **Add to PATH** while installing.



After installing open CMD and verify the installation.

If successful then install, Node-Red using the CMD line

Then start node red, in chrome paste link

Then verify:

</> Bash

```
node -v
npm -v
```



If both work, install Node-RED:

</> Bash

```
npm install -g --unsafe-perm node-red
```



Then start it:

</> Bash

```
node-red
```



Finally open:

```
http://localhost:1880
```



Click on the 3 lines, on the top right corner of the Node – Red

Step 2 — Install Dashboard Package

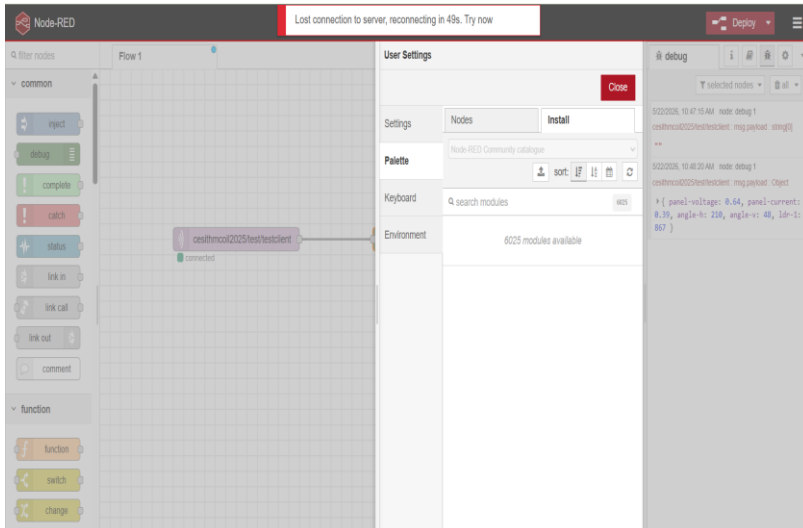
Inside Node-RED:

Top-right menu ☰ → Manage Palette → Install

Search:

node-red-dashboard

Install it.



Then build Flow for the Team .